Predicting Customer Churn Using Machine Learning

Group Number: 6
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***Abstract* — Customer churn prediction is a critical challenge for subscription-based companies, as customer retention is directly correlated with long-term profitability. This project focuses on developing and evaluating machine learning models to predict customer churn using the "Telco Customer Churn" data set. The problem is formulated as a binary classification task, leveraging demographic, behavioral, and service-related features. The project entails comprehensive data preprocessing, exploratory data analysis, and the implementation of multiple algorithms, ranging from baseline models like Logistic Regression, Decision Trees, and K-Nearest Neighbors to advanced techniques such as Random Forest, CatBoost, and Multi-Layer Perceptron. Our final results demonstrate a high-performing ensemble approach that provides actionable insights for customer retention strategies.**

Keywords* — *Customer churn, binary classification, Telco, machine learning, customer retention, predictive modeling.

I. Introduction

Customer churn stands as one of the most significant challenges for subscription-based companies, particularly in industries like telecommunications where revenue depends heavily on long-term customer retention. Companies lose recurring revenue when a client discontinues their subscription, and they simultaneously incur additional costs for onboarding new clients. As a result, customer churn prediction using machine learning has emerged as a critical element in modern business analytics.

This project focuses on predicting customer churn using Telco's Customer Churn dataset [1], [2], which simulates a real-world environment of customer behaviour in the telecommunications industry. This dataset encompasses a wide variety of attributes, including demographic, behavioural, and service-related features. These features introduce significant complexity due to their categorical nature, class imbalance, and non-linear correlation with the target variable.

The primary motivation for this project is to explore how machine learning models translate into real-world application, specifically addressing problems with direct financial implications. Accurate customer churn prediction allows organizations to take proactive measures rather than reactive ones. By recognizing complex attrition patterns, companies can develop target retention strategies, optimize marketing budgets, and ultimately improve overall operational efficiency to retain long-term customers.

The remainder of this paper is organized as follows: Section II presents the background and related work regarding classification techniques. Section III describes the Telco dataset and the exploratory data analysis used to identify key relationships and patterns. Section IV provides the methodology and data preprocessing steps, detailing the implementation of baseline models. Section V discusses the methodology and implementation of advanced machine learning models and ensemble techniques. Section VI presents the experimental results, including a comprehensive evaluation of model performances. Finally, Section VII summarizes the conclusion achieved in this study and discusses potential future research directions.

II. Background and Related Work

Customer churn prediction has been extensively studied within the telecommunication sector due to its direct impact on profitability. Traditional approaches often relied on basic statistical methods, but modern analytics have shifted toward machine learning to capture complex customer behaviors. Recent studies have demonstrated that ensemble methods often outperform single-classifier approaches when dealing with tabular datasets. For instance, research analyzing telecom churn highlights that algorithms like Random Forest and Gradient Boosting consistently achieve higher accuracy and recall than basic Support Vector Machines (SVM) or standalone Decision Trees [3].

Furthermore, contemporary approaches emphasize the need to handle categorical variables and class imbalances efficiently. Advanced models have been introduced to deal with drifting customer variables without requiring extensive pre-processing, treating churn as a dynamic rather than static problem [4]. While deep learning models like Convolutional Neural Networks (CNN) are the standard for image classification, tabular demographic data benefits from algorithms that provide both high predictive power and business interpretability. Building upon this existing body of work, our project combines transparent baseline models (Logistic Regression, Decision Tree, KNN) with advanced ensemble techniques (random Forest, CatBoost, Multi-Layer Perceptron) to achieve robust predictive performance and deliver actionable retention strategies.

III. Exploratory Data Analysis

The exploratory data analysis phase was carried out to understand the structure, quality and underlying patterns within the Telco Customer Churn Dataset before the model development. We used the pandas DataFrame to load the dataset of 7043 customer records and 21 features, including demographic and billing information [1], [2]. Initially, inspection of the dataset involved examining data types, dimensions and missing values. Most columns were correctly formatted, aside from the TotalCharges feature, as it had blank entries that were unable to be correctly processed. So, this column was converted to numeric format, with invalid or missing entries being replaced with 0 to represent customers with no accumulated charges. For the

target variable, churn, we converted it into a binary format where 'Churn' is 1 and 'No Churn' is 0.

A. Class Imbalance and Distribution

To understand the class imbalance, the distribution of the target variable was visualized. The visualizations demonstrated a clear class imbalance, where the majority of customers did not churn: approximately 73.5% of customers were retained, and 26.5% churned (Fig. 1). This imbalance highlights the importance of using evaluation metrics beyond accuracy and justifies the later use of techniques such as class weighting.

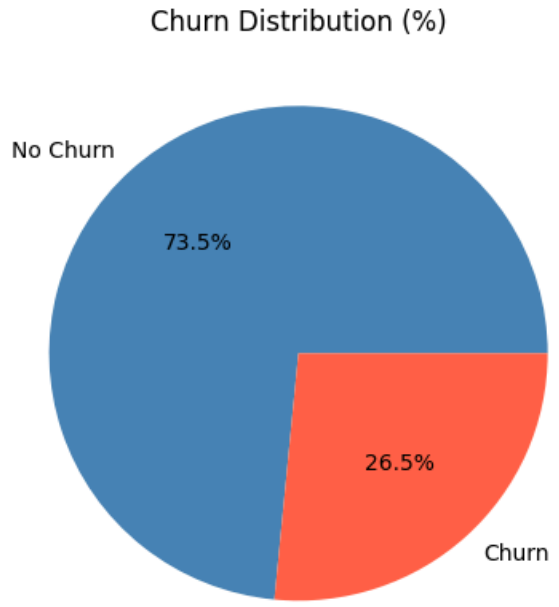


Figure 1. Customer Churn Distribution

B. Tenure Analysis

Customer tenure, measured in months, was analyzed to identify its relationship with churn behavior. It was revealed that customers with shorter tenure are significantly more likely to churn, while long-term customers tend to remain subscribed. A boxplot further emphasized this trend, displaying that the median tenure of churned customers is substantially lower than that of retained customers (Fig. 2). Additionally, an analysis of the financial feature showed how monthly charges affect attrition, indicating a higher churn density among customers with more expensive monthly plans (Fig. 3). These findings and visualizations clearly display that early-stage customer engagement is critical for retention.

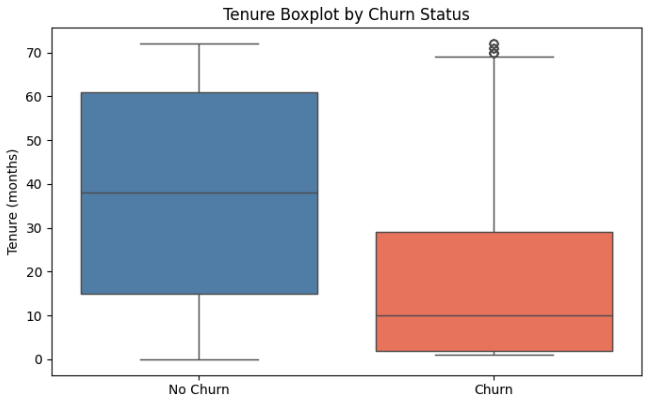


Figure 2. Customer Tenure Box Plot by Churn

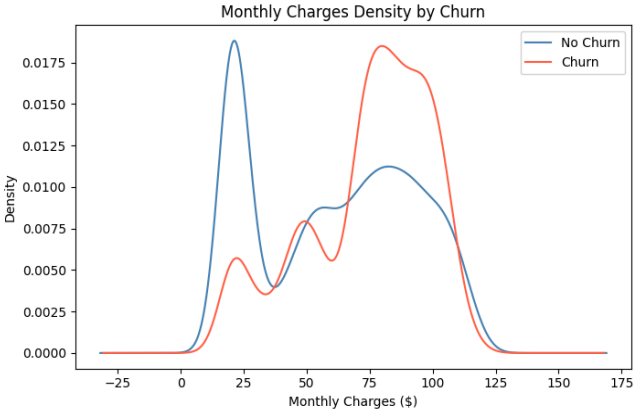


Figure 3. Monthly Charges Density by Churn

C. Contract Type Analysis

The impact of contract type on churn was analyzed to identify service-level trends. Customers on month-to-month exhibit the highest churn rates, while customers with one-year or two-year contracts show significantly lower churn rates. This shows a direct correlation between longer contracts and lower churn rates. When normalized, the percentage of churn is clearly highest among short-term contract users (Fig. 4).

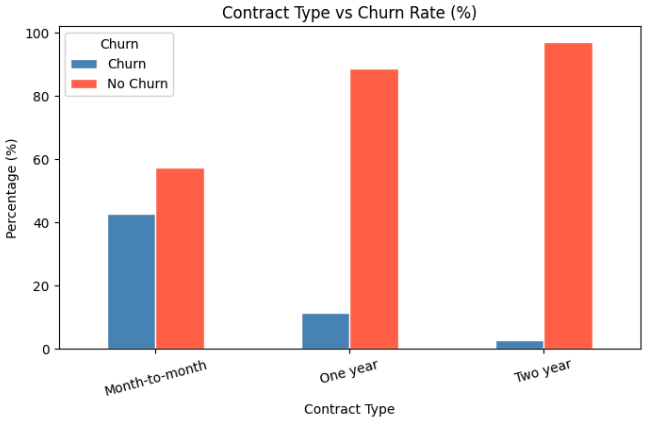


Figure 4. Churn Distribution by Contract Type

D. Key Insights from EDA

The exploratory analysis stage revealed several important patterns. Firstly, a class imbalance exists, with churn being the minority class. Since only 26.5% of customers churned, the model could become biased toward predicting that customers would stay. Secondly, short tenure is strongly correlated with a higher churn risk. When customers do not make long-term commitment, they are far more likely to churn. This displays that the early stages in customer engagement are crucial for retention. It is also worth noting that, paired with short tenure, higher monthly charges are also associated with higher churn. When a customer has an expensive, month-to-month contract type, they are more likely to churn compared to a cheaper, yearly contract. These findings guided our feature selection and modeling strategies, particularly in handling class imbalance and emphasizing the more critical variables.

IV. Methodology of Baseline Models

In this section we describe the implementation of three baseline machine learning models used to establish a performance benchmark for predicting customer churn:

Logistic Regression, Decision Trees, and K-Nearest Neighbors (KNN).

A. Data Cleaning and Preprocessing

Before applying machine learning models, a rigorous data cleaning phase was conducted to ensure the integrity of the Telco dataset. First, we identified blank strings in the TotalCharges column, which were replaced with '0' to present new customers with no accumulated charges, allowing a consistent numerical conversion. Additionally, the customerID column was dropped as it contained unique identifiers with no predictive power. The target variable, Churn, was encoded into a binary format, mapping 'Yes' to 1 and 'No' to 0.

Following the cleaning, the categorical features were transformed using one-hot encoding, resulting in a feature space of 30 variables. The dataset was then partitioned using an 80/20 stratified train-test split to preserve the original churn ratio. Finally, a StandardScaler was applied to a training set to normalize numerical features, ensuring that distance-based algorithms like KNN do not weigh features with larger scales more heavily.

B. Logistic Regression

Logistic Regression was implemented as a primary statistical baseline due to its efficiency and well-understood probabilistic output for binary classification problems. Theoretically, it models the probability of a default class (Churn) by passing a linear combination of input features through a sigmoid (logistic) function, mapping output between 0 and 1. To effectively handle the severe class imbalance in the Telco dataset (where approximately 26.5% of customers churn), we utilized the class_weight='balanced' parameter. This technique adjusts the loss function's weights inversely proportional to class frequencies, forcing the algorithm to penalize classification errors on the minority churn class more heavily.

C. Decision Trees

A Decision Tree classifier was trained to capture potential non-linear decision boundaries without assuming a linear relationship among features. The algorithm works by recursively partitioning the dataset into subsets based on features thresholds that maximise information gain or minimize Gini impurity, ultimately assigning a class label at the leaf nodes. Like the logistic model, it was configured with balanced class weights. This model was selected for its high interpretability, providing a transparent structure to eventually visualize how specific features and contract types split the data to influence the final churn prediction.

D. K-Nearest Neighbors (KNN)

The KNN algorithm was employed with n = 5 neighbors as a non-parametric baseline. It operates on the premise of feature similarity, it classifies the new data point based on the majority vote of its k nearest neighbors in the standardized feature space, relying on Euclidean distance. It serves as a valuable benchmark to compare against the parametric linear approach of Logistic Regression and the ruled-based approach of Decision Tree, providing a model-free perspective on the dataset's structure.

V. Methodology of Advanced Models

To improve upon the baseline benchmarks, three advanced machine learning models were implemented. Random Forest, CatBoost, and a Multi-Layer Perceptron (MLP). These models were selected because their ability to capture nonlinear relationships, model complex feature interactions, and handle class imbalance are something linear classifiers cannot.

A. Hyperparameter Tuning Strategy

All three advanced models were tuned using RandomizedSearchCV with a StratifiedKFold cross-validation strategy of five folds. To preserve the 26.5%/73.5% churn ratio across each fold, Stratified splitting was applied. ROC-AUC was selected as the optimization metric as it is robust to class imbalance and evaluates model discrimination across all classification thresholds. RandomizedSearchCV was preferred over exhaustive GridSearchCV because it samples a fixed number of parameter combinations. Thirty for Random Forest and twenty for CatBoost and MLP which achieves near-optimal performance at a significantly lower computational cost.

B. Random Forest

Random forest constructs a large number of decision trees during training and is an ensemble learning method. The decision trees are each built on a random subset of the data and features. The final prediction is determined by majority vote across all trees. The bagging approach reduces the variance associated with individual decision trees and also mitigates overfitting. To address the dataset's class imbalance, class_weight='balanced' was applied. It adjusts each tree's loss function inversely proportional to class frequencies. The hyperparameters search space included the number of estimators, minimum samples per split and leaf, maximum tree depth, and the maximum number of features considered at each split.

C. CatBoost

CatBoost is a gradient boosting algorithm which was developed by Yandex that builds an ensemble of decision trees sequentially. Each tree corrects the residual of its predecessor. One key advantage of CatBoost for this dataset is its native handling of categorical features through ordered target statistics, which eliminated the need for manual one-hot encoding prior to training. Class imbalance was handled through the scale_pos_weight parameter and was set to the ratio of non-churners to churners in the training set. The tuned hyperparameters included the number of boosting iterations, tree depth, L2 regularization strength, learning rate, and subsampling ratio.

D. Multi-Layer Perceptron (MLP)

The MLP is a neural network that is feedforward. It consists of an input layer, one or more hidden layers, and an output layer. Each neuron applies a weighted linear combination of its inputs followed by a nonlinear activation function. Either ReLU or tanh, which enables the network to approximate complex nonlinear relationships between features and the target variable. The network is trained via backpropagation, continuously updating weights to minimize the binary cross entropy loss. To prevent overfitting L2 regularization and early stopping with a validation fraction of 10% were applied. The StandardScaler

that was applied during preprocessing was critical here. The gradient descent converges significantly faster and with more reliability on normalized features. The tuned parameters included activation function, learning rate, regularization strength, layer architecture, and maximum iterations.

VI. Experimental Results and Evaluation

This section details the experimental results and the evaluation of the implemented machine learning models, establishing a performance benchmark on the Telco Customers Churn dataset.

A. Baseline Models Performance Analysis

The performance of the baseline models was evaluated using a comprehensive suite of metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The results (Table I and Fig. 5), indicate that Logistic Regression achieved the highest Recall (78.61%), which is critical for retention strategies where identifying as many potential churners as possible is the priority. Furthermore, Logistic regression achieved the highest ROC-AUC score (84.14%), indicating it possesses the strongest overall ability to distinguish between churning and retaining customers across different probability thresholds (Fig. 6). To further validate these results in the context of an unbalanced dataset, the Precision-Recall curve (Fig. 7) was analyzed; it confirms that Logistic Regression maintains a superior balance between capturing churners and minimizing false alarms compared to its baseline competitors. While KNN and Decision Tree reached similar baseline Accuracy (~73-74%), their lower F1-Score and Recall demonstrate that Accuracy alone is misleading metric for imbalance datasets, confirming Logistic regression as the most robust baseline model.

TABLE I. Performance Metrics for Baseline Models

Model	Logistic Regression	Decision Tree	KNN
Accuracy (%)	74.02	73.10	74.73
Precision (%)	50.69	49.31	52.53
Recall (%)	78.61	47.59	50.00
F1-Score (%)	61.64	48.44	51.23
ROC-AUC (%)	84.14	65.00	77.16

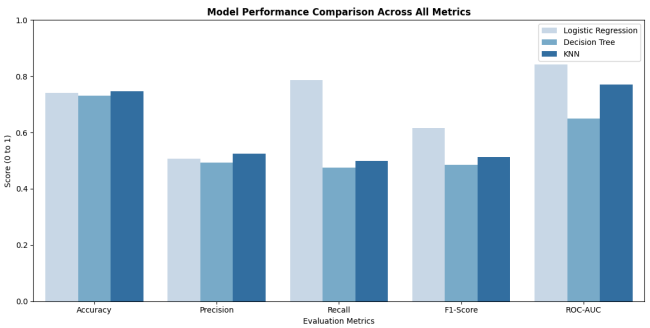


Figure 5. Metrics Comparison Bar Chart

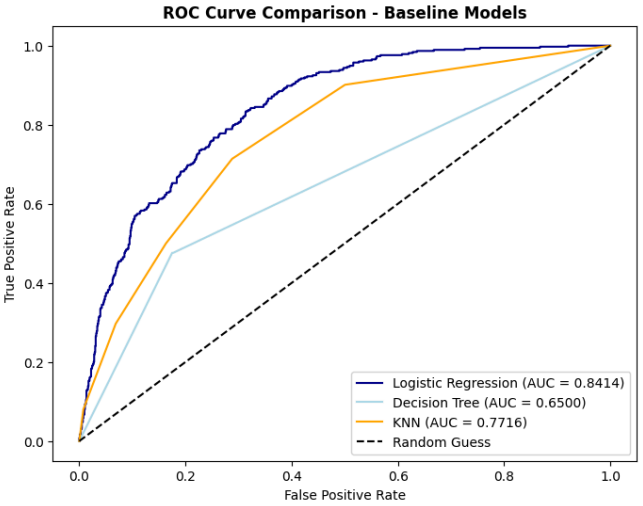


Figure 6. ROC Curve Comparison

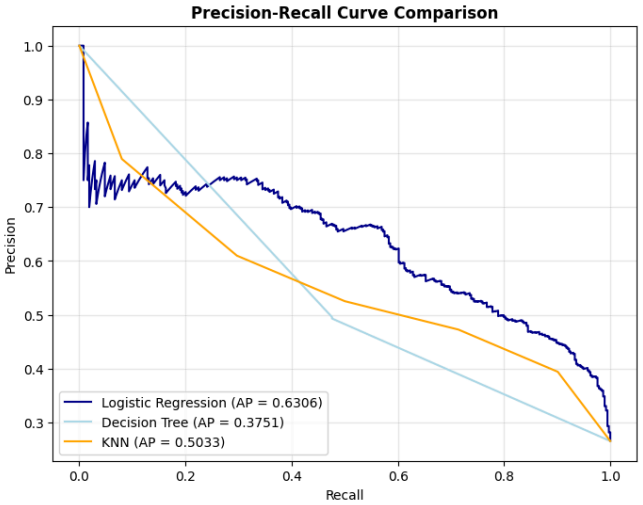


Figure 7. Precision-Recall Curve

B. Baseline Models Observation

As shown in the confusion matrix (Fig. 8, Fig. 9, Fig. 10), the models exhibit distinct behaviors in handling the dataset's class imbalance. Logistic Regression demonstrates a superior ability to correctly identify churners, successfully capturing 294 True positives. This effectiveness drives its high Recall (78.61%), although it comes at the cost of a higher number of False Positives (286) compared to the other models, resulting in a moderate Precision of 50.69%. In contrast, the Decision Tree shows a lower capability to identify the minority class, with a near-even split between False Negatives (196) and True Positives (178), leading to a much lower Recall of 47.59%. This indicates signs of lower generalization on this specific feature set, further evidenced by its low ROC-AUC score of 65.00%. Finally, the K-Nearest Neighbors (KNN) model demonstrates a strong classification of the majority class (Retained, with 866 True Positives) but struggles to capture the minority Churn class effectively, yielding an equal number of True Positives and False Negatives (187) and a Recall of exactly 50.00%. Ultimately, while KNN favors the majority class, Logistic regression proves to be the most practical baseline choice for a business context where minimizing missed churners (False Negatives) is the strategic priority. Finally, taking advantage of the Decision Tree's interpretability, a feature importance analysis was conducted (Fig. 11). The results highlight the most influential variables in the dataset,

providing actionable business insights for future retention strategies.

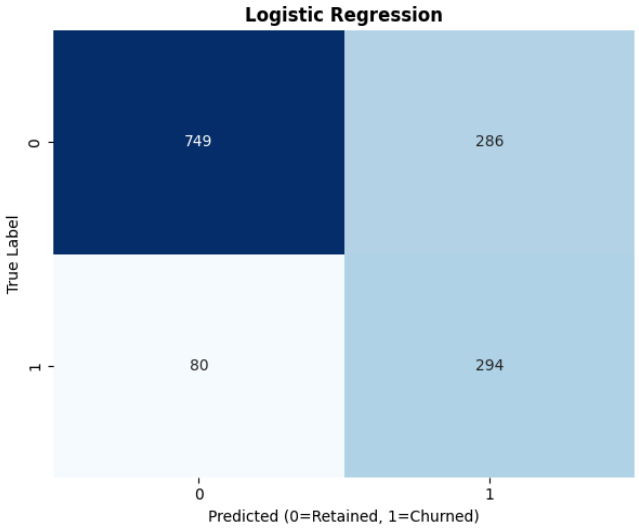


Figure 8. Confusion Matrix: Logistic Regression

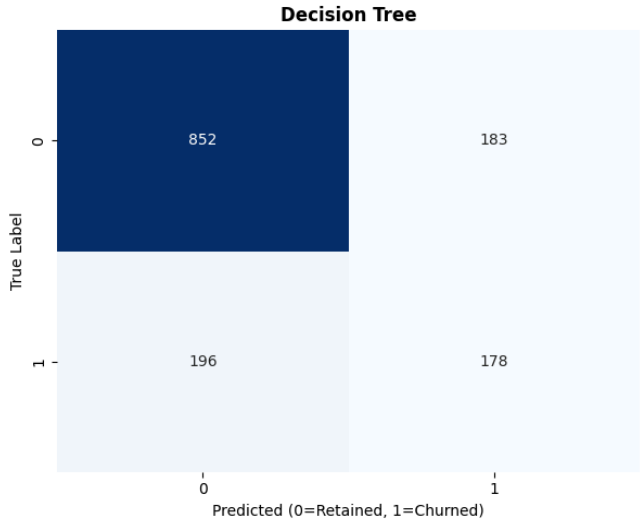


Figure 9. Confusion Matrix: Decision Tree

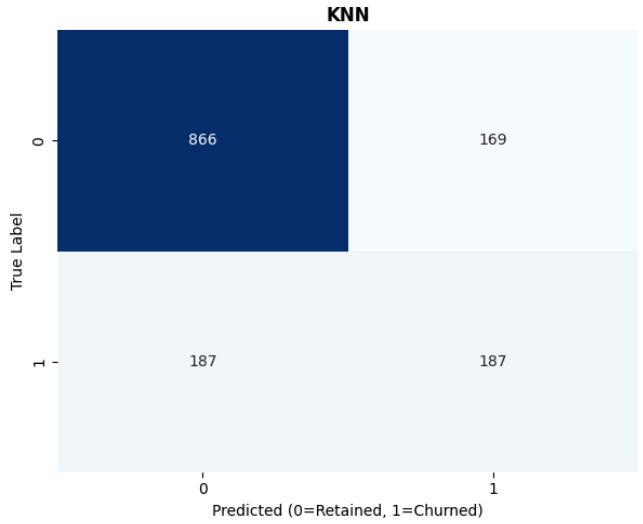


Figure 10. Confusion Matrix: K-Nearest Neighbors (KNN)

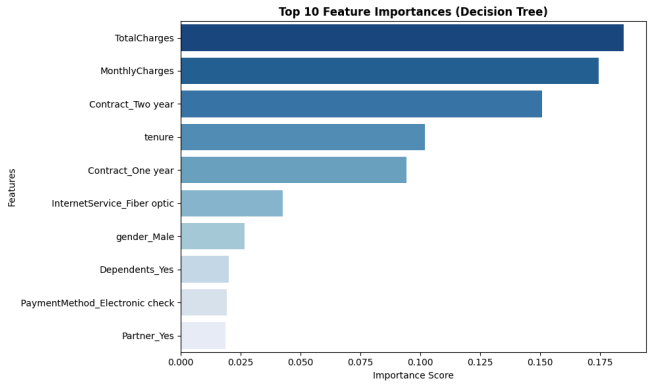


Figure 11. Top 10 Feature Importances extracted from Decision Tree

C. Advanced Models Performance Analysis

The three advanced models, Random Forest, Catboost, and MLP, were evaluated on the same test set using the same metrics as the baseline models (Table II).

Catboost achieved the highest ROC-AUC score of 84.72%, surpassing Random Forest (84.44%) and Logistic Regression (84.14%), and demonstrated the highest Recall of 80.75% among all six models. This proves Catboost is the most effective model for identifying at-risk customers, as it minimizes missed churners (False negatives). Random Forest was the one that achieved a strong balance across metric with an F1 Score of 63.29% and Recall of 76.74%, outperforming all baseline models on the F1 metric. The MLP achieved the highest overall Accuracy (79.56%) and Precision (63.87%) which indicated a stronger tendency to avoid false alarms, at a cost of a lower Recall (52.94%). Furthermore, the ROC curves illustrate that all three advanced models curve substantially above the random baseline with Catboost and Random forest closely tracking each other throughout and outperforming the MLP at lower false positive rates (Fig. 12).

Comparing the advanced models to the baseline, the most significant improvements are seen in Recall and F1-Score. CatBoost's Recall of 80.75% exceeds Logistics Regressions 78.61%. Which makes it superior for the churn detection objective while also achieving a higher ROC-AUC. Random Forests F1-Score of 63.29% surpasses all baselines which demonstrates that ensemble methods better handle the nonlinear structure of this dataset, which is consistent with findings in related literature [3].

TABLE II. PERFORMANCE METRICS FOR ADVANCED MODELS

Model	Random Forest	CatBoost	MLP
Accuracy (%)	76.37	74.02	79.56
Precision (%)	53.85	50.67	63.87
Recall (%)	76.74	80.75	52.94
F1-Score (%)	63.29	62.27	57.89
ROC-AUC (%)	84.44	84.72	83.81

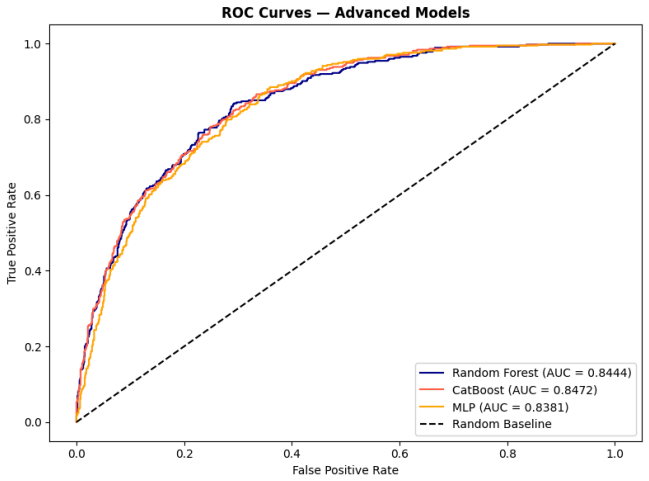


Figure 12. Advanced Models ROC Curves

D. Advanced Models Observation

To interpret the Random Forest model's predictions, Mean Decrease in Impurity (MDI) feature importance scores were extracted for the top 15 features (Fig. 13). Tenure emerged as the single most influential predictor with a MDI score of approximately 0.175. It was followed by TotalCharges and Contract_Two year. InternetService_Fiber and MonthlyCharges also ranked high, consistent with the patterns identified during exploratory data analysis. Notably, the two-year contract and online security subscription both appeared among the top predictors. They reinforce the finding that contractual commitment and value-added services are key drivers of customer retention. These results align with the business insight that early-tenure, month-to-month, and high charge customers represent the highest risk cohort and should be the primary focus of targeted retention strategies.

To further validate and interpret the CatBoost model at the individual prediction level, SHAP (SHapley Additive Explanations) values were computed using a TreeExplainer, ranking the top 15 features by their mean absolute SHAP value (Fig. 14). Each dot represents a single customer colored by feature value from low (blue) to high (red). Tenure is confirmed as the dominant predictor: customers with low tenure values (blue dots) cluster on the right side of the plot, which indicates they push the model strongly toward predicting churn, whereas high-tenure customers (red) push towards retention. Contract_Two year shows a strong negative SHAP effect, meaning customers on two year contracts consistently reduce churn probability, directly supporting the EDA findings. InternetService_Fiber optic and PaymentMethod_Electronic check both show a positive SHAP contribution, indicating these features increase churn risk. TotalCharges exhibits a wide distribution of SHAP values which reflects its interaction with tenure; customers with low total charges tend to be a newer and higher risk. OnlineSecurity_Yes and TechSupport_Yes notably both show negative SHAP values, confirming that the presence of these services acts as a retention factor. Overall, the SHAP analysis corroborates the Random Forest feature importance results while also revealing the direction of each feature's influence, providing a more complex and actionable picture of the drivers behind customer churn.

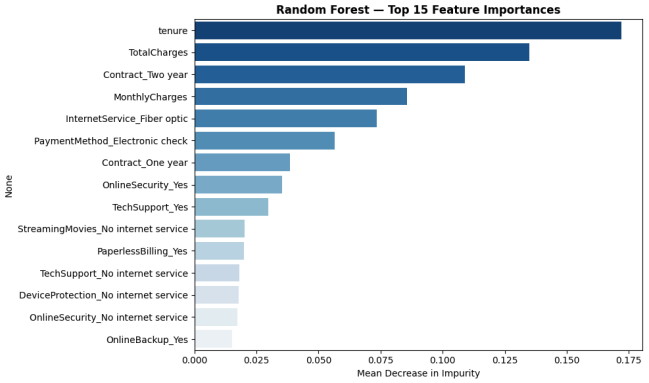


Figure 13. Random Forest - Top 15 Feature Importances

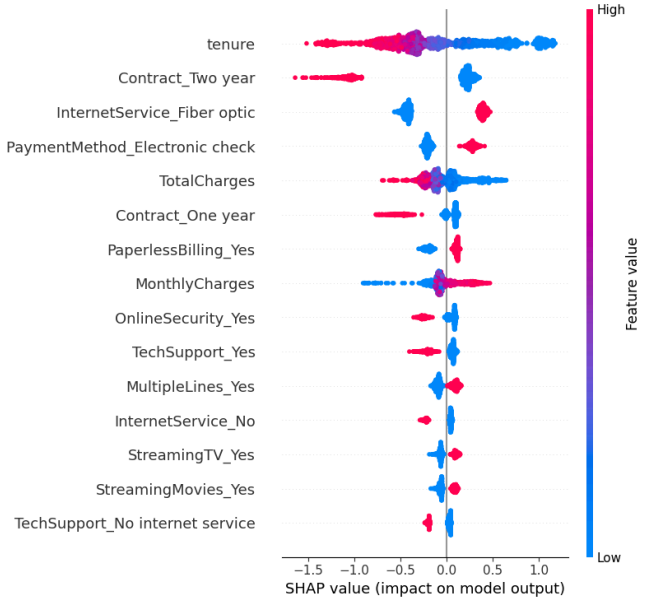


Figure 14. Catboost TreeExplainer Results

VII. Conclusion

This project demonstrated the effectiveness of machine learning techniques in predicting customer churn within the telecommunications industry. By evaluating both baseline and advanced models, it was observed that ensemble methods including Random Forest and CatBoost provided superior predictive performance, while Logistic Regression remained a strong and interpretable baseline, specifically in the identification of at-risk customers through its high recall. This project also uncovered critical drivers of churn, like short customer tenure, high monthly charges, and contract types that used month-to-month billing. These findings bring valuable insight into customer behavior, and have highlighted the importance of early customer engagement and long-term contract incentives. Overall, this project not only achieved accurate predictive results, but also demonstrated how machine learning can generate actionable business insights, reinforcing its value as a tool for data-driven decision making in real world applications.

VIII. Contribution Statement

Alexander Younis: Responsible for the data cleaning and preprocessing phase, including handling missing values and features encoding.

Yusuf Mohamed: Conducted the exploratory data analysis (EDA), identifying the key trends and patterns within the Telco dataset.

Gabriele Martorano: Implemented the baseline machine learning models (Logistic Regression, Decision Trees and KNN) and performed the initial performance evaluation.

Adyan Iqbal: Responsible for the implementation and tuning of advanced models (Random Forest, CatBoost and MLP) and the analysis of feature importance.

All authors: contributed for drafting, technical documentation, and final review of the project report.

Deliverables

- Group6 - Project - FinalCode: https://colab.research.google.com/drive/1EP0PAdI_VAsd_WRHg3-wnvPERnd7cnEY?usp=sharing

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